Suggested Answers to Problem Set 4

1. Given that

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1},$$

$\alpha_0 > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, and $\alpha_1 + \beta_1 < 1$, we can derive the ARCH representation by inverting the lag polynomial:

$$\begin{aligned} h_t &= (1 - \beta_1 L)^{-1} \left(\alpha_0 + \alpha_1 \varepsilon_{t-1}^2\right) \\ &= \left(1 + \beta_1 L + \beta_1^2 L^2 + \beta_1^3 L^3 + \ldots\right) \left(\alpha_0 + \alpha_1 \varepsilon_{t-1}^2\right) \\ &= \left(1 + \beta_1 + \beta_1^2 + \beta_1^3 + \ldots\right) \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_1 \beta_1 \varepsilon_{t-1}^2 + \alpha_1 \beta_1^2 \varepsilon_{t-2}^2 + \alpha_1 \beta_1^3 \varepsilon_{t-3}^2 + \ldots \\ &= \frac{\alpha_0}{1 - \beta_1} + \alpha_1 \sum_{i=1}^{\infty} \beta_1^{i-1} \varepsilon_{t-i}^2. \end{aligned}$$

2. (a)

$$E_{t-1} y_t \equiv E\left(y_t | \Omega_{t-1}\right) = E\left(\phi_0 + \phi_1 y_{t-1} | y_{t-1}, y_{t-2}, \ldots\right) = \phi_0 + \phi_1 y_{t-1}.$$

$$\begin{aligned} Var\left(y_t | \Omega_{t-1}\right) &= E_{t-1}\left(y_t - E_{t-1} y_t\right)^2 = E_{t-1}\left(y_t - \phi_0 - \phi_1 y_{t-1}\right)^2 \\ &= E_{t-1}\left(\varepsilon_t^2\right) = E\left(h_t v_t^2 | \varepsilon_{t-1}, \varepsilon_{t-2}, \ldots\right) = h_t E\left(v_t^2\right) \\ &= h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2. \end{aligned}$$

(b)

$$E\left(y_t\right) = \phi_0 + \phi_1 E\left(y_{t-1}\right) + E\left(\varepsilon_t\right).$$

Under the assumptions, $E\left(y_t\right) = E\left(y_{t-1}\right)$, $E\left(\varepsilon_t\right) = E\left[E\left(\varepsilon_t | \Omega_{t-1}\right)\right] = E\left[\sqrt{h_t} E\left(v_t | \varepsilon_{t-1}, \varepsilon_{t-2}, \ldots\right)\right] = 0$, we then have

$$E\left(y_t\right) = \frac{\phi_0}{1 - \phi_1}.$$

Since

$$\begin{aligned} Var\left(\varepsilon_t\right) &= E\left(\varepsilon_t^2\right) = E\left[E\left(\varepsilon_t^2 | \Omega_{t-1}\right)\right] = E\left(h_t\right) \\ &= E\left(\alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2\right) \\ &= \alpha_0 + \alpha_1 E\left(\varepsilon_{t-1}^2\right) + \alpha_2 E\left(\varepsilon_{t-2}^2\right) \end{aligned}$$

and the stationarity assumptions hold, we have

$$Var\left(\varepsilon_t\right) = E\left(\varepsilon_t^2\right) = \frac{\alpha_0}{1 - \alpha_1 - \alpha_2}.$$

Then the unconditional variance of y_t is

$$\begin{aligned} Var\left(y_t\right) &= \phi_1^2 Var\left(y_{t-1}\right) + Var\left(\varepsilon_t\right) + 2Cov\left(y_{t-1}, \varepsilon_t\right) \\ &= \phi_1^2 Var\left(y_t\right) + Var\left(\varepsilon_t\right), \end{aligned}$$

and hence

$$Var\left(y_t\right) = \left(\frac{1}{1-\phi_1^2}\right)\left(\frac{\alpha_0}{1-\alpha_1-\alpha_2}\right).$$

3. (a) Let $\Omega_{t-1} = \{\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots\}$, and note that h_{t-1} depends on $\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots$. Then

$$\begin{aligned} E\left(\varepsilon_t^2|\Omega_{t-1}\right) &= E\left(h_t v_t^2|\Omega_{t-1}\right) = h_t E\left(v_t^2|\Omega_{t-1}\right) = h_t \\ &= \alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \beta_1 h_{t-1}. \end{aligned}$$

(b) First, $E\left(\varepsilon_t\right) = E\left[E\left(\varepsilon_t|\Omega_{t-1}\right)\right] = E\left[\sqrt{h_t}E\left(v_t|\Omega_{t-1}\right)\right] = 0$. Then $Var\left(\varepsilon_t\right) = E\left(\varepsilon_t^2\right) = E\left[E\left(\varepsilon_t^2|\Omega_{t-1}\right)\right]$. Using the result in (a), we have

$$Var\left(\varepsilon_t\right) = E\left(h_t\right) = E\left(\alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \beta_1 h_{t-1}\right),$$

or

$$E\left(h_t\right) = \alpha_0 + \alpha_1 E\left(\varepsilon_{t-1}^2\right) + \alpha_2 E\left(\varepsilon_{t-2}^2\right) + \beta_1 E\left(h_{t-1}\right).$$

Under the stationarity assumption, $E\left(\varepsilon_{t-1}^2\right) = E\left(\varepsilon_{t-2}^2\right) = E\left(\varepsilon_t^2\right) = E\left(h_t\right) = E\left(h_{t-1}\right)$. Therefore,

$$Var\left(\varepsilon_t\right) = E\left(h_t\right) = \frac{\alpha_0}{1-\alpha_1-\alpha_2-\beta_1}.$$

(c) For $h = 1$,

$$\hat{h}_{t+1|t} = E\left(h_{t+1}|\Omega_t\right) = \alpha_0 + \alpha_1\varepsilon_t^2 + \alpha_2\varepsilon_{t-1}^2 + \beta_1 h_t.$$

For $h = 2$,

$$\begin{aligned} \hat{h}_{t+2|t} &= E\left(h_{t+2}|\Omega_t\right) = \alpha_0 + \alpha_1 E\left(\varepsilon_{t+1}^2|\Omega_t\right) + \alpha_2\varepsilon_t^2 + \beta_1 E\left(h_{t+1}|\Omega_t\right) \\ &= \alpha_0 + \alpha_1 E\left(h_{t+1}v_{t+1}^2|\Omega_t\right) + \alpha_2\varepsilon_t^2 + \beta_1\hat{h}_{t+1|t} \\ &= \alpha_0 + \alpha_1\hat{h}_{t+1|t}E\left(v_{t+1}^2|\Omega_t\right) + \alpha_2\varepsilon_t^2 + \beta_1\hat{h}_{t+1|t} \\ &= \alpha_0 + \left(\alpha_1+\beta_1\right)\hat{h}_{t+1|t} + \alpha_2\varepsilon_t^2. \end{aligned}$$

In general, for $h > 2$,

$$\begin{aligned} \hat{h}_{t+h|t} &= E\left(h_{t+h}|\Omega_t\right) = \alpha_0 + \alpha_1 E\left(\varepsilon_{t+h-1}^2|\Omega_t\right) + \alpha_2 E\left(\varepsilon_{t+h-2}^2|\Omega_t\right) + \beta_1 E\left(h_{t+h-1}|\Omega_t\right) \\ &= \alpha_0 + \alpha_1\hat{h}_{t+h-1|t} + \alpha_2\hat{h}_{t+h-2|t} + \beta_1\hat{h}_{t+h-1|t} \\ &= \alpha_0 + \left(\alpha_1+\beta_1\right)\hat{h}_{t+h-1|t} + \alpha_2\hat{h}_{t+h-2|t}. \end{aligned}$$

4.

(a) We need to solve

$$|\mathbf{\Phi} - \lambda \mathbf{I}_k| = det\left[\begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right] = det\begin{pmatrix} 0.2-\lambda & 0.3 \\ -0.6 & 1.1-\lambda \end{pmatrix} = 0,$$

which is equivalent to

$$(0.2-\lambda)(1.1-\lambda) - 0.3 \times (-0.6) = 0,$$

$$\lambda^2 - 1.3\lambda + 0.4 = 0.$$

The two roots are $\lambda_1 = 0.5$ and $\lambda_2 = 0.8$. Both lie inside the unit circle ($|\lambda| < 1$). Therefore, $\mathbf{y}_t$ is weakly stationary.

(b) Let $\boldsymbol{\mu} = E(\mathbf{y}_t)$. Since $\mathbf{y}_t$ is stationary, we have

$$\boldsymbol{\mu} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix} \boldsymbol{\mu},$$

and it follows that

$$\begin{aligned} \boldsymbol{\mu} &= \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}\right]^{-1} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 0.8 & -0.3 \\ 0.6 & -0.1 \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 3 \\ -6 & 8 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -8 \end{pmatrix}. \end{aligned}$$

(c) The 1-step-ahead forecast at origin T is given by

$$\hat{\mathbf{y}}_{T+1|T} = \hat{\boldsymbol{\phi}}_0 + \hat{\mathbf{\Phi}}\mathbf{y}_T = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 4.35 \\ 1.95 \end{pmatrix}.$$

That is, $\hat{y}_{1,101} = 4.35$ and $\hat{y}_{2,101} = 1.95$. The forecast error is

$$\boldsymbol{e}_{T+1|T} = \mathbf{y}_{T+1} - \hat{\mathbf{y}}_{T+1|T} = \boldsymbol{\varepsilon}_T,$$

which has a covariance matrix $\mathbf{\Sigma}$. Therefore, the standard errors of $\hat{y}_{1,101}$ and $\hat{y}_{2,101}$ are given by $\sigma_1 = \sqrt{1.0} = 1$ and $\sigma_2 = \sqrt{1.5} = 1.22$, respectively.

(d) Write the MA representation as

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t + \mathbf{\Psi}_1\boldsymbol{\varepsilon}_{t-1} + \mathbf{\Psi}_2\boldsymbol{\varepsilon}_{t-2} + \ldots = \boldsymbol{\mu} + \sum_{i=0}^{\infty} \mathbf{\Psi}_i\boldsymbol{\varepsilon}_{t-i}.$$

Denote

$$\mathbf{\Phi} = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}, \quad \mathbf{\Phi}(L) = \mathbf{I}_2 - \mathbf{\Phi} L.$$

Then we have

$$\begin{aligned} \mathbf{y}_t &= \mathbf{\Phi}(L)^{-1} \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \mathbf{\Phi}(L)^{-1} \boldsymbol{\varepsilon}_t \\ &= \boldsymbol{\mu} + \left(\mathbf{I}_2 + \mathbf{\Phi} L + \mathbf{\Phi}^2 L^2 + \ldots\right) \boldsymbol{\varepsilon}_t \\ &= \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t + \mathbf{\Phi}\boldsymbol{\varepsilon}_{t-1} + \mathbf{\Phi}^2 \boldsymbol{\varepsilon}_{t-2} + \ldots. \end{aligned}$$

You can also verify this result by recursive substitution. Therefore, we obtain

$$\begin{aligned} &\mathbf{\Psi}_0 = \mathbf{I}_2, \\ &\mathbf{\Psi}_1 = \mathbf{\Phi} = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}, \\ &\mathbf{\Psi}_2 = \mathbf{\Phi}^2 = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}^2 = \begin{pmatrix} -0.14 & 0.39 \\ -0.78 & 1.03 \end{pmatrix}. \end{aligned}$$

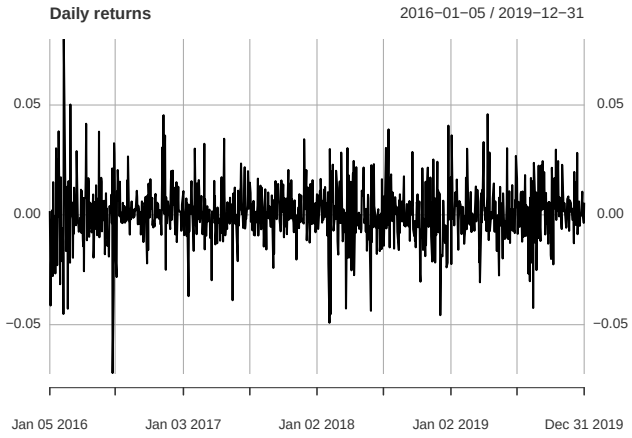
Question 5

(a) The return series r_t clearly exhibits volatility clustering: large changes in prices tend to cluster together.

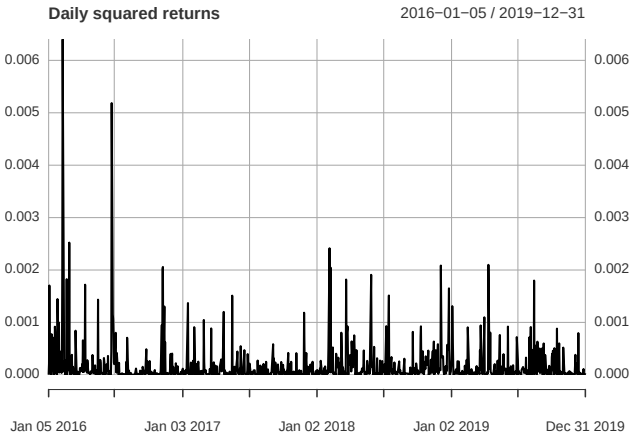
(b) Both the ACF plot and Ljung-Box test result suggest that there is no serial correlation in r_t. Both the Ljung-Box and Lagrange multiplier tests applied to r_t^2 yield p-values close to zero. We find strong evidence of ARCH effects in r_t.

```
library(quantmod)
library(forecast)
library(rugarch)

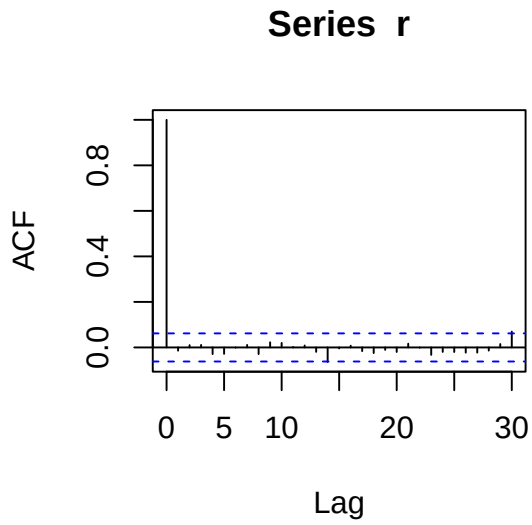
JPM_full <- getSymbols("JPM", src="yahoo", auto.assign = F)
JPM <- JPM_full["2016/2019"]
r <- diff(log(JPM$JPM.Close))[-1]
plot(r, main="Daily returns")
```

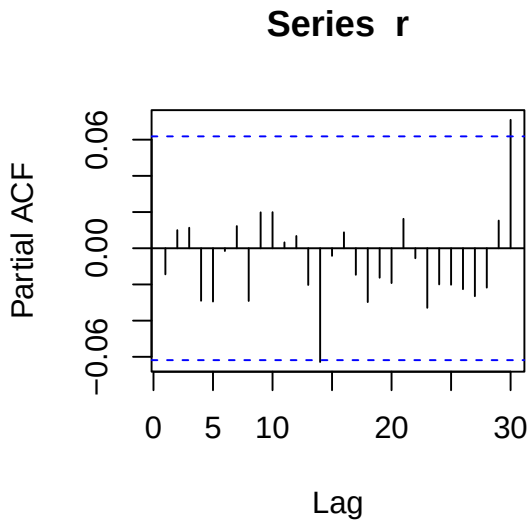



```
plot(r^2, main="Daily squared returns")
```




```
par(mfrow = c(1,2))
acf(r)
pacf(r)
```

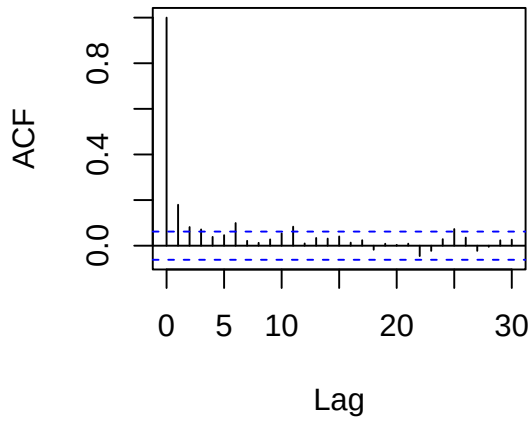


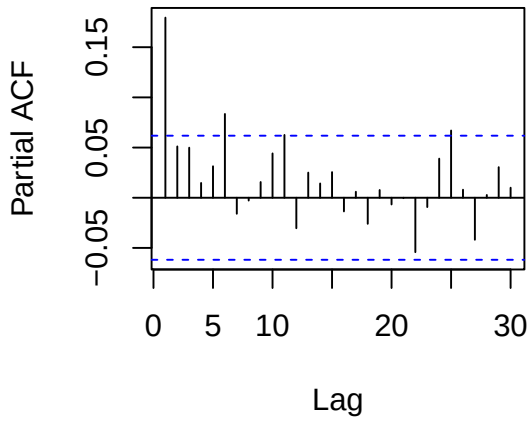



```
Box.test(r, lag = 10, type="Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  r
## X-squared = 4.0214, df = 10, p-value = 0.9464
```

```
acf(r^2)
pacf(r^2)
```






```
Box.test(r^2, lag = 10, type="Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  r^2
## X-squared = 62.344, df = 10, p-value = 1.302e-09
```

```
library(FinTS)
ArchTest(r, 10)  # Engle's ARCH LM test
```

```
##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  r
## Chi-squared = 47.967, df = 10, p-value = 6.295e-07
```

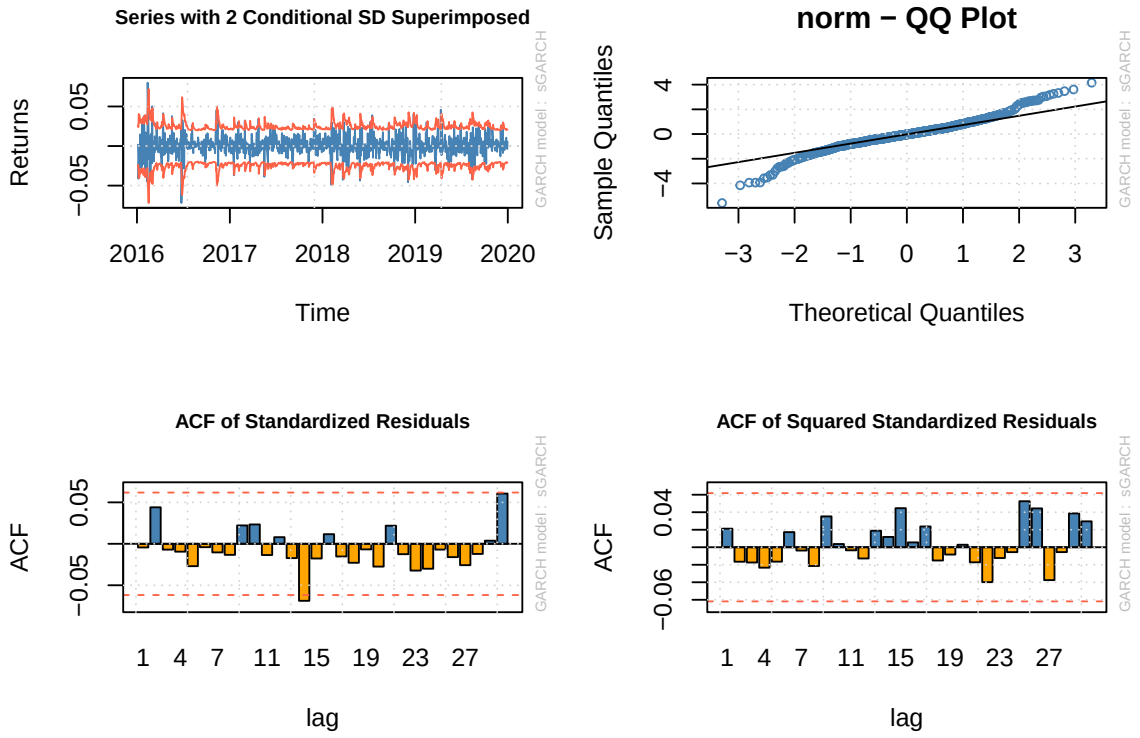
(c) The (Weighted) Ljung-Box tests show that both the standardized residuals and standardized squared residuals are serially uncorrelated, which is further confirmed by the ACF plots. However, the QQ plot and the Pearson Goodness-of-Fit tests, which compare the empirical distribution of the standardized residuals with the theoretical density, suggest that the normality assumption is not valid. Therefore, this model appears inadequate.

```
spec_garch11 <- ugarchspec(variance.model = list(garchOrder=c(1,1)),
                           mean.model = list(armaOrder=c(0,0)))
fit_garch11 <- ugarchfit(spec=spec_garch11, data=r)
fit_garch11
```

```
##
## *---------------------------------*
## *          GARCH Model Fit        *
## *---------------------------------*
##
## Conditional Variance Dynamics
## -----------------------------------
## GARCH Model  : sGARCH(1,1)
## Mean Model   : ARFIMA(0,0,0)
## Distribution : norm
##
## Optimal Parameters
## ------------------------------------
##         Estimate  Std. Error  t value Pr(>|t|)
## mu      0.000878    0.000376   2.3351 0.019540
## omega   0.000035    0.000011   3.1978 0.001385
## alpha1  0.152639    0.040994   3.7234 0.000197
## beta1   0.642470    0.091998   6.9835 0.000000
##
## Robust Standard Errors:
##         Estimate  Std. Error  t value Pr(>|t|)
## mu      0.000878    0.000425   2.0643 0.038986
## omega   0.000035    0.000022   1.5785 0.114458
## alpha1  0.152639    0.082993   1.8392 0.065889
## beta1   0.642470    0.187971   3.4179 0.000631
##
## LogLikelihood : 2981.343
##
## Information Criteria
## ------------------------------------
##
## Akaike       -5.9251
## Bayes        -5.9055
## Shibata      -5.9251
## Hannan-Quinn -5.9176
##
## Weighted Ljung-Box Test on Standardized Residuals
## ------------------------------------
```

```
##                           statistic p-value
## Lag[1]                      0.01968  0.8884
## Lag[2*(p+q)+(p+q)-1][2]     0.99765  0.4990
## Lag[4*(p+q)+(p+q)-1][5]     1.79801  0.6669
## d.o.f=0
## H0 : No serial correlation
##
## Weighted Ljung-Box Test on Standardized Squared Residuals
## ------------------------------------
##                           statistic p-value
## Lag[1]                       0.4536  0.5006
## Lag[2*(p+q)+(p+q)-1][5]      1.1336  0.8288
## Lag[4*(p+q)+(p+q)-1][9]      1.8398  0.9232
## d.o.f=2
##
## Weighted ARCH LM Tests
## ------------------------------------
##             Statistic Shape Scale P-Value
## ARCH Lag[3]    0.3054 0.500 2.000  0.5805
## ARCH Lag[5]    0.9065 1.440 1.667  0.7609
## ARCH Lag[7]    1.1465 2.315 1.543  0.8886
##
## Nyblom stability test
## ------------------------------------
## Joint Statistic:  0.8366
## Individual Statistics:
## mu     0.05104
## omega  0.09062
## alpha1 0.38777
## beta1  0.15903
##
## Asymptotic Critical Values (10% 5% 1%)
## Joint Statistic:          1.07 1.24 1.6
## Individual Statistic:     0.35 0.47 0.75
##
## Sign Bias Test
## ------------------------------------
##                    t-value   prob sig
## Sign Bias           0.4857 0.6273
## Negative Sign Bias  1.2549 0.2098
## Positive Sign Bias  0.5896 0.5556
## Joint Effect        2.2825 0.5159
##
##
## Adjusted Pearson Goodness-of-Fit Test:
## ------------------------------------
##   group statistic p-value(g-1)
## 1    20     70.04    9.042e-08
## 2    30     76.52    3.695e-06
## 3    40     90.44    5.756e-06
## 4    50    104.80    6.229e-06
##
##
## Elapsed time : 0.2602341
```

```
par(mfrow = c(2,2))
plot(fit_garch11, which=1)
plot(fit_garch11, which=9)
plot(fit_garch11, which=10)
plot(fit_garch11, which=11)
```



(d) The GARCH(1,1) model with standardized Student-t innovations seems adequate. The residuals of the mean and variance equations are serially uncorrelated. Moreover, the QQ plot and the Pearson Goodness-of-Fit tests indicate that the empirical quantiles of the standardized residuals align well with the theoretical quantiles. The fitted model is given by

$$
\begin{aligned}
r_t &= 6.61 \times 10^{-4} + \varepsilon_t, \\
\varepsilon_t &= \sqrt{h_t} v_t, \quad v_t \sim IIDt^*_{3.7}, \\
h_t &= 1.9 \times 10^{-5} + 0.145 \varepsilon^2_{t-1} + 0.766 h_{t-1},
\end{aligned}
$$

where $t^*_{3.7}$ denotes the standardized t distribution with 3.7 degrees of freedom.

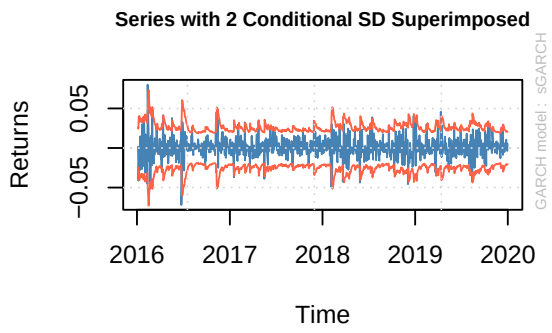
```
spec_garch11std <- ugarchspec(variance.model = list(garchOrder=c(1,1)),
                              mean.model = list(armaOrder=c(0,0)),
                              distribution.model = "std")
fit_garch11std <- ugarchfit(spec=spec_garch11std, data=r)
fit_garch11std
```

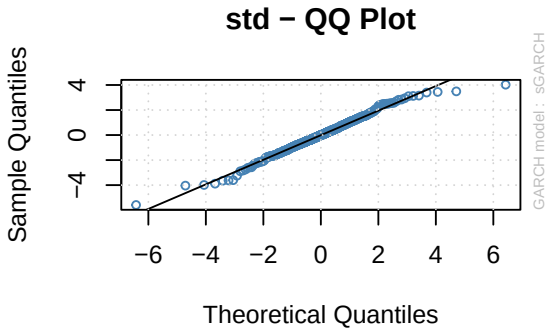
```
##
## *---------------------------------*
## *          GARCH Model Fit        *
## *---------------------------------*
##
## Conditional Variance Dynamics
## -----------------------------------
## GARCH Model  : sGARCH(1,1)
```

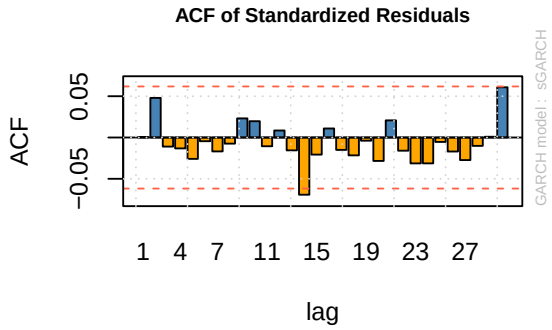
```
## Mean Model   : ARFIMA(0,0,0)
## Distribution : std
##
## Optimal Parameters
## ------------------------------------
##         Estimate  Std. Error  t value Pr(>|t|)
## mu      0.000661    0.000314   2.1040 0.035380
## omega   0.000019    0.000008   2.3362 0.019480
## alpha1  0.145371    0.046107   3.1529 0.001616
## beta1   0.766014    0.069600  11.0059 0.000000
## shape   3.702635    0.467101   7.9268 0.000000
##
## Robust Standard Errors:
##         Estimate  Std. Error  t value Pr(>|t|)
## mu      0.000661    0.000301   2.1951 0.028157
## omega   0.000019    0.000010   1.8591 0.063011
## alpha1  0.145371    0.053511   2.7167 0.006594
## beta1   0.766014    0.088059   8.6989 0.000000
## shape   3.702635    0.403662   9.1726 0.000000
##
## LogLikelihood : 3044.303
##
## Information Criteria
## ------------------------------------
##
## Akaike       -6.0484
## Bayes        -6.0239
## Shibata      -6.0484
## Hannan-Quinn -6.0391
##
## Weighted Ljung-Box Test on Standardized Residuals
## ------------------------------------
##                         statistic p-value
## Lag[1]                  0.0001753  0.9894
## Lag[2*(p+q)+(p+q)-1][2] 1.1611397  0.4488
## Lag[4*(p+q)+(p+q)-1][5] 2.1372011  0.5864
## d.o.f=0
## H0 : No serial correlation
##
## Weighted Ljung-Box Test on Standardized Squared Residuals
## ------------------------------------
##                         statistic p-value
## Lag[1]                     0.7134  0.3983
## Lag[2*(p+q)+(p+q)-1][5]    1.8314  0.6588
## Lag[4*(p+q)+(p+q)-1][9]    2.9973  0.7594
## d.o.f=2
##
## Weighted ARCH LM Tests
## ------------------------------------
##             Statistic Shape Scale P-Value
## ARCH Lag[3]    0.4414 0.500 2.000  0.5065
## ARCH Lag[5]    1.9912 1.440 1.667  0.4732
## ARCH Lag[7]    2.4051 2.315 1.543  0.6320
##
```

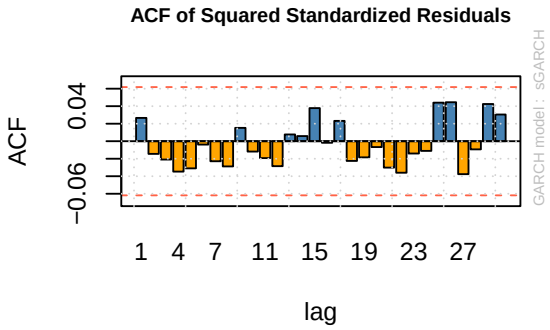
```
## Nyblom stability test
## ------------------------------------
## Joint Statistic:  0.8286
## Individual Statistics:
## mu     0.07489
## omega  0.20247
## alpha1 0.15104
## beta1  0.22851
## shape  0.31407
##
## Asymptotic Critical Values (10% 5% 1%)
## Joint Statistic:          1.28 1.47 1.88
## Individual Statistic:     0.35 0.47 0.75
##
## Sign Bias Test
## ------------------------------------
##                    t-value   prob sig
## Sign Bias           0.7781 0.4367
## Negative Sign Bias  1.4567 0.1455
## Positive Sign Bias  0.7615 0.4465
## Joint Effect        2.9085 0.4059
##
##
## Adjusted Pearson Goodness-of-Fit Test:
## ------------------------------------
##   group statistic p-value(g-1)
## 1    20     8.433       0.9818
## 2    30    14.552       0.9882
## 3    40    34.403       0.6794
## 4    50    44.801       0.6439
##
##
## Elapsed time : 0.2723749
```

```
par(mfrow = c(2,2))
plot(fit_garch11std, which=1)
plot(fit_garch11std, which=9)
plot(fit_garch11std, which=10)
plot(fit_garch11std, which=11)
```







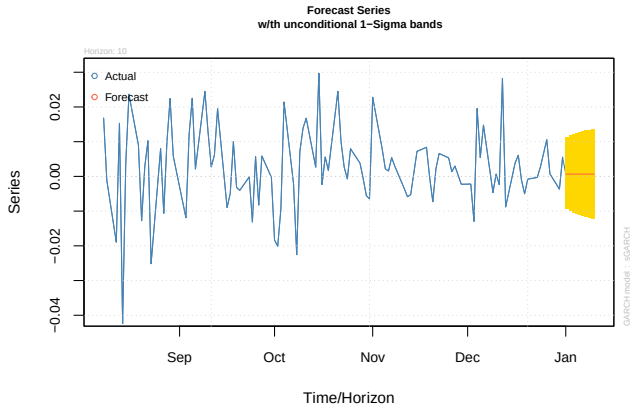


(e) The forecasts of the returns and conditional standard deviation are given below.

```
(fcst_garch11std <- ugarchforecast(fit_garch11std, n.ahead=10))
```

```
##
## *----------------------------------*
## *       GARCH Model Forecast       *
## *----------------------------------*
## Model: sGARCH
## Horizon: 10
## Roll Steps: 0
## Out of Sample: 0
##
## 0-roll forecast [T0=2019-12-31]:
##         Series   Sigma
## T+1  0.0006611 0.01012
## T+2  0.0006611 0.01062
## T+3  0.0006611 0.01105
## T+4  0.0006611 0.01143
## T+5  0.0006611 0.01176
## T+6  0.0006611 0.01206
## T+7  0.0006611 0.01232
## T+8  0.0006611 0.01256
## T+9  0.0006611 0.01277
## T+10 0.0006611 0.01296
```

```
par(mfrow = c(1, 1))
plot(fcst_garch11std, which=1)
```




```
plot(fcst_garch11std, which=3)
```

